Reduce RTO without cancelling growth

A risk-based COD policy for the next operating pilot

TO Founder	FROM Associate Software Engineer Candidate	DATE 30 Oct 2026

RECOMMENDATION

Do not block COD by pincode or automatically cancel a first order. Use the model to route low-risk orders straight to dispatch, ask mid-risk customers to confirm, offer a ₹25 prepaid incentive to higher-risk orders, and hold only the very highest-risk orders for a person to review. Automatic cancellation stays off until a controlled pilot proves that genuine-order loss is acceptably low.

MODEL RTO SCORE	ACTION	WHY
<20%	Ship as-is	Low predicted risk
20% to <30%	WhatsApp confirmation	Positive expected value above ~19.2%
30% to <70%	Offer ₹25 prepaid discount	Convert a higher-risk COD order
≥70%	Hold for manual review	Do not dispatch before a person checks it

WHAT THE SUPPLIED ORDERS SHOW

- **Clean baseline.** After removing 30 exact duplicate rows and one order still marked IN_TRANSIT, 1,987 labelled orders remain: 565 RTOs and 1,422 delivered. RTO is 28.4%, close to the brief's 28% assumption.
- **Repeat RTO history is the clearest customer signal.** Orders with two prior RTOs returned 26 of 34 times (76.5%); one prior RTO returned 38 of 108 (35.2%); zero prior RTOs returned 501 of 1,845 (27.2%). The two-RTO group is small, so use it to trigger review, not an automatic block.
- **Geography varies, but should not become a blacklist.** Bihar recorded 53.8% RTO (78/145) and Uttar Pradesh 44.1% (86/195), versus Maharashtra at 18.2% (70/385). These are descriptive correlations; courier mix and sample size can explain part of the gap.
- **Acquisition channel matters to the trade-off.** Meta Ads orders returned at 31.2% (239/765) versus 22.9% for Direct (58/253). Marketing reports ₹300+ CAC; cancelling a first-time Meta order can waste that acquisition spend. First-time buyers are therefore never auto-cancelled by this pilot.
- **The sample and brief disagree on basket size.** The cleaned sample averages ₹920 (median ₹748), while the brief states ₹649 AOV. All business calculations below use the brief's ₹649 so the forecast is not inflated by the sample mix. Three zero-value baskets are treated as missing model inputs; four rows report two prior RTOs despite one prior order, so those source values are flagged for validation before live use.

Economics, validation and guardrails

For validation, we trained on 1,658 April-August orders and tested on 329 September orders (86 RTO; 26.1%). Holdout metrics were ROC-AUC 0.749, PR-AUC 0.567 and Brier 0.163. Iterations were selected using August before September scoring; the saved app model was then refit on all labelled April-September data. This is useful ranking power, not a production guarantee.

WHAT THE SEPTEMBER REPLAY WOULD HAVE DONE

ACTION BUCKET	ORDERS	OBSERVED RTO	RTO RATE
Ship as-is	88	8	9.1%
Confirm first	125	27	21.6%
Prepaid nudge	113	48	42.5%
Hold / manual review	3	3	100%

₹ TRADE-OFF: USE THE BRIEF'S ECONOMICS

- **Cost of one RTO.** ₹65 forward + ₹65 return + ₹25 packaging/handling + (8% × ₹180 damage) = ₹169.40 expected cost.
- **Value of a delivered order.** ₹649 AOV × 62% gross margin = ₹402.38 contribution. CAC is already spent, so it is not subtracted twice; the 50% prevention / 5% timeout assumptions break even at 19.2% RTO risk, so confirmation starts at 20%.
- **Prepaid incentive.** A ₹25 discount is 3.85% of AOV. The RTO gap from 28% COD to 4% prepaid avoids 24% × ₹169.40 = ₹40.66 per converted order; after the discount, that leaves ₹15.66 before gateway fees. The discount breaks even once COD RTO exceeds about 18.8%.

SEPTEMBER SCENARIO (329 ORDERS)	ESTIMATED VALUE
Expected RTO cost avoided	₹5,105
Gross contribution at risk from genuine cancels	(₹2,261)
Expected prepaid discount cost	(₹848)
Net impact vs shipping all COD orders	+₹1,997

Important: This is a scenario, not measured savings. It assumes 50% confirmation prevention, 5% genuine-order timeout, 30% prepaid conversion and that the three highest-risk orders are ultimately cancelled after human review. The operating policy does not auto-cancel. Validate these assumptions in a controlled pilot.

HOW THIS BALANCES OPERATIONS AND MARKETING

- **Operations gets a filter, not a blanket ban.** Use reversible confirmation/prepaid steps and human review for high-risk orders; no location alone can trigger cancellation.
- **Marketing keeps genuine new customers.** First-order or Meta status is not a penalty. Use confirmation or a ₹25 incentive before risking ₹402.38 contribution; treat the ₹300+ CAC as sunk, not a recoverable shipping cost.

WHAT COULD GO WRONG / HOW WE WILL NOTICE

- **Drift or misleading geography/courier patterns.** Track weekly predicted-versus-actual RTO by state, courier, acquisition source and new/repeat status; alert when calibration or false-positive rate deteriorates.
- **Confirmation friction loses real buyers.** Measure response rate, timeouts, cancellation rate and delivered conversion by cohort; reduce reminder friction or adjust thresholds if genuine customers abandon.
- **LLM misreads a Hinglish address or unfamiliar name.** Keep the LLM optional, require consent, use structured output, show its reasons to an agent, and never allow it to cancel an order.

NEXT TWO WEEKS

Connect Shopify and WhatsApp sandbox events; A/B-test the flow; reconcile AOV and courier/pincode data; recalibrate thresholds. Keep auto-cancel off until measured contribution is positive and genuine-order loss is acceptable.